

IV. DRAFT QUESTIONNAIRE

A. GENERAL INFORMATION ABOUT THE RESPONDANT:

• **Country*:** Uzbekistan

Organization(s) or entity(ies) responsible for the preparation of the report:

Name: Agency for Innovative Development under the Ministry of Higher Education, Science and Innovations of the Republic of Uzbekistan

Website: <https://innovation.gov.uz/en>

Email address*:

Phone number*:

Please describe the role/mandate of your organization: Carries out practical measures in the field of innovative and scientific and technological development of the Republic of Uzbekistan, aimed at the comprehensive development of public and state life, increasing the intellectual and technological potential of the country.

Evaluates innovation activities based on indicators of its effectiveness, determines the main directions for the development of relevant industries and areas that require the primary introduction of advanced technologies.

Coordinates the activities of government bodies, research and information and analytical institutions and other organizations on the implementation of innovative technologies, ideas and R&D.

Acts as a sole customer of state scientific and technical programs and projects conducted by scientific, educational and other institutions.

Introduction of scientific achievements in priority areas and industries as well as developing the country's long-term innovative development strategies.

Increasing the intellectual and technological potential of the regions, the formation of modern infrastructures for scientific and innovative activities.

Development of proposals for the formation of state programs related to scientific and innovative activities.

Coordinates the training system of the staff with a scientific degree.

Introduction of effective mechanisms for involving young people in scientific and innovative activities and providing comprehensive support for their innovative initiatives.

Attracting investments, increasing the activity of the private sector and developing venture financing in the implementation of scientific and innovative projects.

Conducting financial, economic and technical expertise of projects implemented on the basis of innovative developments and providing the necessary conditions for the implementation of intellectual property objects.

Determination of the needs of the real sector of the economy and industrial sectors in scientific developments and innovations, as well as strengthening the integration of science and production based on the development of sectoral research works.

Assistance in the commercialization and introduction of new developments into the industry, the implementation of startup projects, the creation of the necessary conditions for the development of rationalization activities.

Establishing and strengthening of international relations in the field of science, innovation and technology transfer.

• **Officially designated contact person(s) who completed this survey: Full name*:**
Farkhod Kudratov (Mr)

Position*: Chief specialist

Email address*:

Full Name(s) of designated official(s) certifying the report: Umidjon Khoshimov (Mr)

• Brief description of the consultation process established for the preparation of the report:

The report was prepared by the Ministry of Higher Education, Science and Innovations, the Agency for Innovative Development, and affiliated organizations, with input from key stakeholders.

• Other organization(s) or entity(ies) (including non-governmental) consulted for completing this survey:

Name of the organization*: Ministry of Higher Education, Science and Innovations, Center for Scientific and Technical Information

Website*: <https://gov.uz/oz/edu>,

Email address*:

Sector: Public

B. Questionnaire for Member States' reporting on the implementation of the 2021 Recommendation

1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE, ASSOCIATED BENEFITS, AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN SCIENCE

1.1 Has the 2021 Recommendation been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

Yes, the 2021 Recommendation has been promoted and shared with appropriate ministries, institutions, and affiliated organizations in our country through the following channels:

Ministry of Higher Education, Science and Innovations – This ministry is responsible for policy development and coordination of scientific and innovation-related initiatives. It has played a key role in sharing and promoting the Open Science Recommendation as part of its ongoing efforts to foster scientific progress.

Agency for Innovative Development – Actively involved in implementing and promoting innovative strategies and policies, the Agency has incorporated aspects of Open Science into its projects and initiatives, contributing to national awareness and implementation.

Center for Scientific and Technical Information – This center plays a pivotal role in the dissemination of scientific and technical advancements. It has supported awareness-building campaigns and educational programs that highlight the importance of Open Science and its benefits.

Other Affiliated Research Institutions and Universities – Various research institutions and universities have engaged in discussions, workshops, and training sessions aimed at integrating the Open Science principles outlined in the 2021 Recommendation into national strategies. These institutions have actively worked to promote the widespread adoption of Open Science practices across their communities.

1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

No, the 2021 Recommendation on Open Science is not fully available in all national languages of our country at this moment. However, a Russian version of the document is currently available. There are also plans to translate it into Uzbek in the near future.

1.3 Have there been awareness raising activities on open science, including all its key elements', associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? (ref.: i 16 16. 16.h

Yes, several awareness-raising activities on Open Science, including its key elements, associated benefits, and challenges, have been organized or are planned to be organized by national authorities and entities by the end of 2025. These activities include:

Workshops and Training Sessions – Organized by the Ministry of Higher Education, Science and Innovations, along with the Agency for Innovative Development, these sessions aim to educate researchers, academics, and policymakers about Open Science principles, its benefits, and strategies for implementation.

National Conferences and Forums – These events are designed to foster discussions among scientists, universities, and research institutions on Open Science practices, challenges, and opportunities, helping to build a collaborative approach toward Open Science.

Publications and Online Resources – Informational materials and digital platforms are being developed to raise awareness of Open Science among a broader audience, including resources that highlight its importance and the practical steps involved in its adoption.

Collaboration with International Organizations – Engagement in partnerships with UNESCO and other international bodies to align national Open Science policies with global best practices, ensuring that the country remains connected to ongoing international efforts in promoting Open Science.

1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open science", in publicly funded research? (ref.: i 16.a. 16.by 16.c 16.d 16.e. 1_Ifi_}.h

YES

In accordance with paragraph 14 of Appendix 3 to the Law of the Republic of Uzbekistan No. URQ-1011, dated December 24, 2024, expenditures for financing projects within the framework of state programs for scientific activity in 2025 are set at 570 billion soums.

Furthermore, paragraphs 19-20 of Appendix 1 to the Decree of the President of the Republic of Uzbekistan No. PF-16, dated January 30, 2025, outline the implementation of fundamental research and the scaling up of projects in a wide range of fields, including mathematics, biology, chemistry, geology, physics, epidemiology, seismology, biodiversity, and engineering. The decree also covers social and humanitarian sciences such as language and literature, history, archaeology, spirituality, culture and art.

These actions will contribute to incorporating the principles of Open Science by fostering research that is accessible, transparent and collaborative.

These projects are planned to be financed from the state budget, utilizing funds allocated for state programs dedicated to scientific activity.

2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

2.1 Does your country have a national policy", strategy or plan of action on science, technology and innovation (STI)?

Yes.

Title of the policy: Decree of the President of the Republic of Uzbekistan No. PF-165 "On approval of the Strategy for the Innovative Development of the Republic of Uzbekistan for 2022 — 2026."

The authority in charge of policy development: Ministry of Higher Education, Science, and Innovations of the Republic of Uzbekistan.

Entities involved: Ministry and administrative bodies, the Council of Ministers of the Republic of Karakalpakstan, regional governments, and the Tashkent city government.

Year of adoption: July 6, 2022.

Key areas/sections of the policy: The strategy prioritizes the following sectors in its implementation:

Supporting startup initiatives and organizing large-scale production by forming a network of innovative infrastructure entities, such as innovative technology parks, technology transfer centers, innovation clusters, venture organizations, innovation centers, startup accelerators, and incubators.

Increasing the share of innovatively active organizations by improving institutional mechanisms for state support of innovative activities.

Ensuring rapid socio-economic growth of regions by increasing the innovative activity of small businesses.

Stimulating demand for innovation by establishing a comprehensive system for creating new types of products and innovative technologies, from the initial idea to the final consumer.

Creating a system for redirecting capital towards "radically renewing" innovations.

Further developing human capital in the management of innovative activities by fostering creativity, innovative entrepreneurship, and rationalization skills at all stages of education.

<https://lex.uz/docs/-6102462>

2.2 In your country, is there a policy, or a set of policies and/or legal framework(s) at the national level that address open science or any of its key elements in line with the definition, values, and principles outlined in the 2021 Recommendation on Open Science? (ref.: ti 16.a 16b 16.c 16.d. 16.p. 6.h)

PARTLY

Decree of the President of the Republic of Uzbekistan No. PF-165, dated 06.07.2022, 'On the Approval of the Strategy for Innovative Development of the Republic of Uzbekistan for 2022–2026.

2.3 In your country, are there specific policy instruments that aim to promote open science?

UNDER DEVELOPMENT

As of now, Uzbekistan does not have specific policy instruments explicitly designed to promote Open Science. However, the broader national strategies and policies related to science, technology, and innovation may support the principles of Open Science in an indirect manner.

For example, Uzbekistan's Strategy of Innovative Development (2022-2026), which focuses on advancing scientific research and innovation, encourages collaboration, knowledge sharing, and innovation infrastructure development. These areas align with Open Science values such as openness, collaboration, and accessible scientific knowledge.

If yes or under development, please provide details and links as relevant, including for each instrument: Title of the instrument, its objective, the authority in charge, allocated budget, elements of open science included.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policy instruments.

2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? (ref.: i ii. 17.a 17.b, 17.c 17.f, 17.h)

YES / UNDER DEVELOPMENT / NO

If yes or under development, please provide details and links as relevant, including the name of ministries and national authorities/entities, as well as affiliated organizations that are involved in the development and implementation of the plan, strategy or the roadmap.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policy instruments.

2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at institutional level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? (ref.: (ii) 17.d, 1 e 1 .

YES /UNDER DEVELOPMENT/ NO

If yes, please provide more information, including the number of policies, name of the institute(s) that has/have developed the policy(s) and elements of open science Error! Bookmark not defined. included in the policy (please consider policies and strategies addressing all elements of open science, including citizen and participatory science, or co-production of knowledge with multiple actors).

2.6 In your country, are there specific funding mechanisms ' for open science at the national level? (ref.: (ii) 17 j, (iii) 18.j, (iv) 19c).

YES.

A specific funding mechanism for open science is established through the Decree of the Cabinet of Ministers of the Republic of Uzbekistan No. 133, dated 09.03.2020, titled "On Measures to Further Improve the Regulatory and Legal Framework for the Development of Scientific Research and Innovation Activities."

<https://www.lex.uz/uz/docs/-4759202>

2.7 In your country, have open science practices been integrated into existing research funding mechanisms?

YES

Decree of the Cabinet of Ministers of the Republic of Uzbekistan No. 133, dated 09.03.2020, On Measures to Further Improve the Regulatory and Legal Framework for the Development of Scientific Research and Innovation Activities.

<https://www.lex.uz/uz/docs/-4759202>

2.8 Have efforts been made or are foreseen to be undertaken by the end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (ref.: ii) 7i).

YES

In this regard, a number of measures and regulatory legal acts have been adopted in Uzbekistan. Specifically, in accordance with Appendix 1a to the Decree of the President of the Republic of Uzbekistan No. PF-6097, dated October 29, 2020, it is planned to increase the share of funds allocated by the private sector for research and development in the total funding of science to 20 percent by 2025. Additionally, in accordance with Paragraph 8 of Appendix 2 to the Decree of the President of the Republic of Uzbekistan No. PF-165, dated July 6, 2022, the volume of private sector investments in development and research is set to reach 6.0 billion soums.

Furthermore, in accordance with Paragraph 27 of Appendix 3 to the Decree of the President of the Republic of Uzbekistan No. PF-16, dated January 30, 2025, "On the State Program for the Implementation of the 'Uzbekistan-2030' Strategy in the Year of Environmental Protection and 'Green Economy,'" the volume of private sector investments in science and innovation is planned to increase to 6.0 billion soums by 2025.

2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy? (ref.: (ii) 17.j, 23.c)

YES.

A national monitoring framework for open science will be implemented in accordance with Resolution No. 133 of the Cabinet of Ministers of the Republic of Uzbekistan, dated March 9, 2020, titled "On Measures to Further Improve the Regulatory and Legal Framework for the Development of Scientific Research and Innovative Activities."

<https://lex.uz/ru/docs/4759202?ONDATE=06.08.2022%2000>

3. INVESTING IN OPEN SCIENCE INFRASTRUCTURE AND SERVICE

3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: iii 8.a)

In accordance with Appendix 2 of the Decree of the President of the Republic of Uzbekistan No. PF-6097, dated October 29, 2020, the task of gradually increasing the funds allocated from the state budget to the science sector to 2 percent of gross domestic product (GDP) by 2030, including 0.8 percent of GDP by 2022, has been set.

Organizational measures have been taken to implement the task, including the diversification of science and technology, as outlined in Resolution No. PP-307 'On Organizational Measures to Implement the Strategy of Innovative Development of the Republic of Uzbekistan for 2022-2026.' In 2022, a total of 5 trillion 853 billion 974 million soums were allocated to science, which accounted for 0.8 percent of gross domestic product (GDP). This included 1 trillion 102 billion 9 million soums from the state budget, 2 trillion 537 billion 750 million soums from organizations and industrial enterprises with a state share, 2 trillion 11 billion 474 million soums from the private sector, and 201.85 billion soums from higher education and scientific organizations. This indicator is expected to reach 0.4 percent of GDP in 2024.

Also, in accordance with paragraph 10 of Appendix 2 to the Decree of the President of the Republic of Uzbekistan No. PF-165, dated July 6, 2022, it is planned to increase total spending on research and development to 0.19% of GDP by 2024. According to the Statistics Agency under the President of the Republic of Uzbekistan, the forecast for the volume of gross domestic product in Uzbekistan for 2024 has increased by 6.6% in current prices, reaching 1.14 quadrillion soums.

<https://stat.uz/uz/matbuot-markazi/qo-mita-yangiliklar/58374-2024-yilning-yanvar-sentabr-oylarida-yalpi-ichki-mahsulot-yaim-ning-o-sish-sur-atlari>

In accordance with paragraph 18 of Appendix 3 to the Law of the Republic of Uzbekistan 'On the State Budget of the Republic of Uzbekistan for 2023,' based on primary data, 778 billion 200 million soums were allocated from the State Budget for research and development this year. Of this, 693.5 billion soums were allocated for financing projects within the framework of state programs on scientific activity, and 84.7 billion soums were allocated for the organization and additional equipment of scientific laboratories with new, modern high-tech equipment.

Additionally, this year, total expenditures on research and development carried out by the private sector amounted to 3 trillion 803 billion 416 million soums, bringing the total gross expenditures for research and development to 4 trillion 581 billion 616 million soums.

Taking the above into account, the gross expenses for research and development this year amounted to 0.40 percent of the forecasted GDP for 2024 in the republic.

3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: iii 8.b)

As of early 2024, Uzbekistan's internet penetration rate stood at 83.3%, indicating that approximately 83.3% of the country's population had access to reliable internet services.

In terms of internet connection speeds, as of January 2024, the median fixed-line broadband download speed was 59.07 megabits per second (Mbps), while the median mobile broadband download speed was 26.97 Mbps.

These figures suggest that a significant portion of Uzbekistan's population has access to reliable internet services, with varying connection speeds depending on the type of service.

3.3 Are there national research and education networks (NRENs) viii active in the country? (ref.: (iii) 18c).

YES

Uzbekistan has a National Research and Education Network (NREN), which is actively operating in the country.

Key Information about Uzbekistan's NREN:

Uzbektelecom, the national telecommunications provider, has been instrumental in establishing the infrastructure for national research and education networks. The "Electronic Education" network, launched in collaboration with the Ministry of Higher and Secondary Special Education, connects higher education institutions through high-speed internet and fiber-optic networks.

3.4 Are there national or institutional open science infrastructures in your country? (ref: (iii) 18.d, 18.e, 18.k, 18.l, 18.m)

YES

It will be implemented in accordance with Resolution No. 133 of the Cabinet of Ministers of the Republic of Uzbekistan, dated March 9, 2020, 'On Measures to Further Improve the Regulatory and Legal Framework for the Development of Scientific Research and Innovative Activities.

<https://lex.uz/ru/docs/4759202?ONDATE=06.08.2022%2000>

3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (ref.: (iii) 18.e, 81f).

YES

Uzbekistan participates in the Central Asian Network of Science and Education (CASNET), which facilitates open science infrastructure for research and collaboration in Central Asia.

Target Groups: Universities, research institutions, and academic bodies in Central Asia.

Scope: Supports high-performance computing, cloud services, and data sharing.

Accessibility: Accessible to relevant users in Uzbekistan and other Central Asian countries for open science initiatives and international collaboration.

3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? (ref.: iii 18.)

YES

Uzbekistan is involved in South-South collaborations, such as through CASNET and other regional initiatives, promoting shared open science platforms and infrastructure. These

collaborations help enhance research and data exchange between Uzbekistan and other countries.

4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY AND CAPACITY BUILDING FOR OPEN SCIENCE

4.1 Has systematic capacity building on open science taken place in higher education and/or research-performing institutions in your country? (ref.: i19 a, 19.c)

YES.

Systematic capacity building on open science has taken place in Uzbekistan's higher education and research-performing institutions.

This includes training programs, workshops, and seminars organized by the Ministry of Higher Education, Science and Innovations, Agency for Innovative Development, and universities. These initiatives aim to educate researchers and students on open science principles, data sharing, and collaborative research practices.

4.2 Have capacity building programmes for policy makers on how to integrate core values and principles of open science in STI policies and strategies taken place in your country?

YES

Uzbekistan has undertaken capacity-building programs for policymakers to integrate open science principles into Science, Technology, and Innovation (STI) policies and strategies. For example, the National Training Workshop on Science, Technology, and Innovation (STI) Park Policies and Practices for Sustainable Development aimed to build the capacity of Uzbekistan's policymakers and STI park practitioners in designing and implementing STI park policies and managing STI parks.

Additionally, the Modernizing Uzbekistan's National Innovation System Project (MUNIS), supported by the World Bank, focuses on enhancing the capacity of public research organizations to achieve international standards of research quality and enable the commercialization of research results.

4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? (ref.: (iv) 19.b)

YES

There have been efforts to incorporate open science competencies into higher education research skills curricula in Uzbekistan.

Educational institutions, such as universities and research centers, have begun to integrate open science principles into their programs. This includes training on data sharing, open-access publishing, and collaborative research. Additionally, it is expected that by 2025, more comprehensive frameworks for open science competencies will be fully integrated

into curricula across universities, driven by initiatives from the Ministry of Higher Education, Science and Innovations and related bodies.

4.4 Are open educational resources ^ as defined in the 2019 Recommendation on Open Educational Resources, used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? (ref.: iv 19.d)

YES

Uzbekistan has utilized Open Educational Resources (OERs) as instruments for open science capacity building, aligning with the 2019 UNESCO Recommendation on Open Educational Resources. For instance, the HiEdTec project developed a virtual library of Uzbekistan, offering digital educational resources based on the Moodle Learning Management System. This initiative aimed to modernize higher education in Central Asia through new technologies, providing open access to educational materials.

Additionally, the Digital Readiness Index survey and Capacity Building for Digital Pedagogy in Uzbekistan project includes capacity building for teachers through hybrid courses titled "The Digital Teacher." Key activities encompass an in-person Master Trainers Orientation Workshop, localization and adaptation, online self-paced certified courses, and a scaling-up strategy.

These initiatives reflect Uzbekistan's commitment to integrating open educational resources into its educational framework, thereby enhancing open science competencies.

4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? (ref.: i 19.e)

YES

Universities and research institutes in Uzbekistan have been granted two years of free access to international scientific databases. Consulting and technical assistance for organizing access is provided by the Center for Scientific and Technical Information. Currently, there are four national subscriptions to international databases in Uzbekistan:

Web of Science: 224 organizations have received access for 3 years.

Springer Nature: 210 organizations have received access for 3 years.

Pivot-RP: 224 organizations have received access for 2 years.

On an ongoing basis, events are held across different regions of the republic to disseminate information about the work and capabilities of these international scientific databases. In 2024 alone, 57 webinars, 51 seminars, and 2 forums were held in various regions and cities of the republic with the participation of foreign specialists.

5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

5.1 Have there been any initiatives at the national or institutional level, e.g., by universities and/or research institutions, to engage the public with science and raise awareness of research by the end of 2025? (ref.: iv. 20.b, 20c, 20i)

YES / NO

If yes, please provide more information and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref.: v 20.a. 20.c 20 d)

YES / NO

If yes, please provide details and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

It will be implemented in accordance with Resolution No. 133 of the Cabinet of Ministers of the Republic of Uzbekistan, dated March 9, 2020, 'On Measures to Further Improve the Regulatory and Legal Framework for the Development of Scientific Research and Innovative Activities.

<https://lex.uz/ru/docs/4759202?ONDATE=06.08.2022%2000>

5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? (ref.: v, 20.a 20.b... 20.c. 20.e 20.f. 20..., 2'0h)

YES

Uzbekistan has undertaken initiatives to recognize and reward open science practices, with plans to expand these efforts by the end of 2025.

The Concept of Development of Science until 2030 outlines strategies to enhance the scientific sector, including the introduction of a national rating system to evaluate the effectiveness of scientific and innovative activities. This system aims to improve social partnerships between the state and scientific organizations, fostering a culture of open science.

Additionally, the government plans to increase the share of private sector funding for research and development from 8% to 20% by 2025, and to 30% by 2030. This initiative is expected to incentivize open science practices by providing financial support and recognition to innovative projects.

These measures reflect Uzbekistan's commitment to promoting open science and rewarding practices that align with its values and principles.

<https://president.uz/ru/lists/view/3926>

5.4 Have there been any unintended negative consequences[^] of open science practices reported in your country? (ref.: 20)

YES / NO

If yes, please provide more information and specify if there is a strategy or plan to preventing and mitigating such consequences.

6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES OF THE SCIENTIFIC PROCESS

6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? (ref.: ifi/)21

YES, at the national level. YES, at the institutional level.

YES, at both national and institutional levels.

NO

If yes, please indicate which of the recommended actions they address and provide more information and links as relevant:

- promotion of pre-prints
- exploration of open peer-review practices
- valuing and sharing of negative scientific results and associated data
- participatory methods that value inputs from societal actors, e.g. citizen science, crowdsourcing scientific projects
- development of participatory strategies for identifying the needs of marginalized communities and highlighting socially relevant issues to be incorporated into research agendas.
- promoting open innovation practices.
- Others (with a box to add text)

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (ref.: (vii) 22.a)

Uzbekistan is actively promoting and facilitating international scientific collaborations on open science, aligning with the 2021 Recommendation on Open Science. Key initiatives include:

Bilateral Agreements: In October 2023, Uzbekistan and China signed a Memorandum of Understanding (MoU) to establish an Open Science Framework. This agreement aims to enhance interoperability between the two countries' scientific data platforms, facilitating the open sharing of scientific data and fostering collaborative research efforts.

Engagement with UNESCO: Uzbekistan is actively involved in UNESCO's Open Science initiatives, contributing to the development of international standards and frameworks that promote open science practices globally. This engagement underscores Uzbekistan's commitment to fostering international cooperation in open science.

Participation in International Platforms: Uzbekistan is a participant in the UNESCO Open Science Platform, which supports the implementation of the 2021 Recommendation. This platform provides a collaborative tool for scientists, communities, policymakers, and the public, facilitating open access to scientific knowledge and promoting cross-regional collaboration.

https://english.cas.cn/newsroom/news/202310/t20231017_383902.shtml

<https://core.unesco.org/en/project/3210241051>

7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: (vii) 22.b, 22.e 22.

YES / NO

If yes, please provide links as relevant and more information.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science?

YES

Uzbekistan is actively involved in discussions and initiatives aimed at establishing regional and international funding mechanisms for open science. Key efforts include:

Engagement with UNESCO: Uzbekistan participates in UNESCO's initiatives to promote open science, which includes discussions on establishing funding mechanisms to support open science efforts globally. This involvement aligns with the 2021 Recommendation on Open Science, which encourages countries to maximize the impact of public funding and

establish regional and international funding mechanisms for promoting and strengthening open science.

Collaboration with the Economic Cooperation Organization (ECO): Uzbekistan is a member of the ECO Science Foundation, an intergovernmental organization that aims to promote scientific and technological research among its member states, including Uzbekistan. The foundation undertakes various initiatives to foster scientific excellence and technical expertise within its member countries, which may include discussions on funding mechanisms for open science.

National Initiatives: At the national level, Uzbekistan has undertaken initiatives to enhance its science and technology infrastructure, which may involve discussions on funding mechanisms for open science. For example, the "Science, Technology and Innovation Potential of the Regions of the Republic of Uzbekistan" document outlines plans to study the financing of scientific research and mechanisms for the commercialization of scientific research results.

https://nsp.gov.uz/static/uploads/an_data44f9f09c-ef74-4c55-8e9e-8c244441791b.pdf

8. GENERAL CONSIDERATION

8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

YES/NO

If yes, please provide more information.